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Controlling an Autonomous Vehicle

Abstract

The present invention relates to a computer-implemented method of controlling an autonomous vehicle. The method comprises: receiving current sensor data defining a current scenario; determining a distance between the current scenario and a closest scenario from a database of known scenarios; and controlling the autonomous vehicle by: manoeuvring the autonomous vehicle within a manoeuvring constraint associated with the closest scenario when the distance is below a first threshold; performing a minimal risk manoeuvre when the distance is above a second threshold, wherein the second threshold is greater than the first threshold; and interpolating the manoeuvring constraint associated with the closest scenario when the distance is between the first threshold and the second threshold, and manoeuvring the autonomous vehicle within the interpolated manoeuvring constraint.

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Background/Summary

FIELD

[0001] The subject-matter of the present disclosure relates to a computer a computer-implemented method of controlling an autonomous vehicle, a computer-implemented method of determining a manoeuvring constraint of an autonomous vehicle during a known scenario, and a transitory, or non-transitory, computer-readable medium.

BACKGROUND

[0002] An autonomous vehicle (AV) may capture a plurality of scenarios. Each scenario may be associated with a travel envelope. When the AV encounters a scenario in real time, it is able to traverse within the travel envelope of a matching scenario.

[0003] A problem exists where there are no known scenario matches.

[0004] It is an aim of the present invention to address such problems and improve on the prior art.

SUMMARY

[0005] According to an aspect of the present disclosure, there is provided a computer-implemented method of controlling an autonomous vehicle. The method comprises: receiving current sensor data defining a current scenario; determining a distance between the current scenario and a closest scenario from a database of known scenarios; and controlling the autonomous vehicle by: manoeuvring the autonomous vehicle within a manoeuvring constraint associated with the closest scenario when the distance is below a first threshold; performing a minimal risk manoeuvre when the distance is above a second threshold, wherein the second threshold is greater than the first threshold; and interpolating the manoeuvring constraint associated with the closest scenario when the distance is between the first threshold and the second threshold, and manoeuvring the autonomous vehicle within the interpolated manoeuvring constraint.

[0006] In an embodiment, the current scenario and each known scenario may be a descriptor representing a multivariable distribution from a bottleneck of a variational autoencoder.

[0007] In an embodiment, the determining the distance may comprise calculating a distance using a plurality of weighted matchers.

[0008] In an embodiment, the plurality of weighted matchers may comprise one or more of: a time warping matcher, a Euclidian distance matcher, and a learned matcher.

[0009] In an embodiment, the manoeuvring constraint may comprise a travel envelope constructed in space-time, within which no infractions are predicted to occur.

[0010] In an embodiment, the minimal risk manoeuvre may comprise one or more of an emergency stop and pulling over to a road side.

[0011] According to an aspect of the invention, there is provided a computer-implemented method of determining a manoeuvring constraint of an autonomous vehicle during a known scenario. The method comprises: running a known scenario of the autonomous vehicle as a simulation in a simulator; iteratively running the simulation by applying different dynamic parameters to the autonomous vehicle; and identifying a maximum dynamic parameter from the different dynamic parameters the autonomous vehicle below which no infractions occur during the simulation and above which an infraction occurs during the simulation; and storing, in a constraint database, the

maximum dynamic parameter as a manoeuvring constraint of the autonomous vehicle for the known scenario.

[0012] In an embodiment, the computer-implemented method further comprises: retrieving a descriptor of the known scenario from a database, the descriptor being a multivariate distribution from a bottleneck of a variational autoencoder; and decoding the descriptor using a decoder from the variational autoencoder to generate the known scenario.

[0013] In an embodiment, the iteratively running the simulation may comprise: incrementing and/or decrementing the dynamic parameters; and checking for infractions at each increment and/or decrement.

[0014] In an embodiment, the computer-implemented method may further comprise: continuing the incrementing and/or decrementing until modulation of the dynamic parameters has been exhausted.

[0015] In an embodiment, the incrementing and/or decrementing may comprise incrementing and/or decrementing at a predefined interval.

[0016] In an embodiment, the dynamic parameters may be selected from a list including velocity, acceleration, deceleration, and jerk.

[0017] In an embodiment, the manoeuvring constraint comprises a travel envelope constructed in space-time, within which no infractions are predicted to occur.

[0018] According to a further aspect of the invention, there is provided a transitory, or non-transitory, computer-readable medium having instructions stored thereon that, when executed by a processor, cause the processor to perform the computer-implemented method of any preceding claim.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] The subject-matter of the present disclosure is best described with reference to the accompanying figures, in which:

[0020] FIG. 1 shows a schematic of an AV according to one or more embodiments;

[0021] FIG. 2 shows a block diagram of features used by the methods described herein;

[0022] FIG. 3 shows a block diagram of features used for determining tracking and prediction of objects within a scenario captured by sensors of the AV from FIG. 1;

[0023] FIG. 4 shows a block diagram of features used for determining tracking and prediction of objects within scenes within a simulator;

[0024] FIG. 5 shows a block diagram of features used for training a scenario encoder;

[0025] FIG. 6 shows a block diagram of features used for comparing scenarios using the trained scenario encoder from FIG. 5;

[0026] FIG. 7 shows a block diagram of the scenario comparator shown in FIG. 6;

[0027] FIG. 8 shows a flow chart of a method of determining a manoeuvring constraint according to one or more embodiments; and

[0028] FIG. 9 shows a flow chart of a method of controlling the AV from FIG. 1.

DESCRIPTION OF EMBODIMENTS

[0029] The embodiments described herein are embodied as sets of instructions stored as electronic data in one or more storage media. Specifically, the instructions may be provided on a transitory or non-transitory computer-readable media. When executed by the processor, the processor is configured to perform the various methods described in the following embodiments. In this way, the methods may be computer-implemented methods. In particular, the processor and a storage including the instructions may be incorporated into a vehicle. The vehicle may be an AV.

[0030] Whilst the following embodiments provide specific illustrative examples, those illustrative examples should not be taken as limiting, and the scope of protection is defined by the claims.

Features from specific embodiments may be used in combination with features from other embodiments without extending the subject-matter beyond the content of the present disclosure. [0031] With reference to FIG. 1, an AV 10 may include a plurality of sensors 12. The sensors 12 may be mounted on a roof of the AV 10. The sensors 12 may be communicatively connected to a computer 14. The computer 14 may be onboard the AV 10. The computer 14 may include a processor 16 and a memory 18. The memory may include the non-transitory computer-readable media described above. Alternatively, the non-transitory computer-readable media may be located remotely and may be communicatively linked to the computer 14 via the cloud 20. The computer 14 may be communicatively linked to one or more actuators 22 for control thereof to move the AV 10. The actuators may include, for example, a motor, a braking system, a power steering system, etc.

[0032] The sensors 12 may include various sensor types. Examples of sensor types include LiDAR sensors, RADAR sensors, and cameras. Each sensor type may be referred to as a sensor modality. Each sensor type may record data associated with the sensor modality. For example, the LiDAR sensor may record LiDAR modality data.

[0033] The data may capture various scenes that the AV 10 encounters. For example, a scene may be a visible scene around the AV 10 and may include roads, buildings, weather, objects (e.g. other vehicles, pedestrians, animals, etc.), etc.

[0034] With reference to FIG. 2, a vehicle control system 30 comprises the plurality of sensors 12. The plurality of sensors 12 includes an inertial measurement unit (IMU) 12_1, a camera 12_2, a LiDAR sensor 12_3, and a radar sensor 12_4. The vehicle control system 30 also comprises an odometry module 32, a localisation module 34, a perception module 36, and a tracking and prediction module 38. The system 30 also comprises a high-level instruction module 40, a mission planning module 42, a mission execution module 44, a trajectory planning module 46, and a controller 48.

[0035] The IMU 12_1 may include an accelerometer and a gyroscope, for example.

[0036] The odometry module 32 may be configured to estimate a position of the AV 10 over time relative to a reference point (e.g. a starting point) using inputs from the plurality of sensors 12.

[0037] The localisation module 34 may be configured to estimate a global pose of the AV 10 within its environment using inputs from the plurality of sensors 12.

[0038] The perception module 36 may be configured to generate models of the AV's 10 environment, including objects within the environment. The models may include segmentation maps, bounding boxes, cuboids, depth perception, etc., using inputs from the plurality of sensors 12.

[0039] The tracking and prediction module 38 may be configured to track objects (static and dynamic) within a scene based on the outputs from the odometry module 32, the localisation module 34, and the perception module 36.

[0040] The high-level instructions module 40 may be configured to receive an input from a user regarding a route, e.g. a destination, and output a command to move the AV 10 from the current location to a destination, e.g. from point A to B.

[0041] The mission planning module 42 may include a map, and when in cooperation with the mission planning module 42, may be configured to construct a route that will execute the route from the high-level instructions module 40. The route may be a travel envelope within which the AV 10 must travel to traverse the route successfully.

[0042] The trajectory planning module 46 may be configured to plan a trajectory for the AV 10 that is within the travel envelope defined by the mission execution module 44, taking account of objects' trajectories and positions from the tracking and prediction module, and the position of the AV 10 from the odometry and localisation modules 32, 34.

[0043] The controller 48 may be configured to control one or more actuators of the AV 10 to implement the trajectory from the trajectory planning module 46.

[0044] With reference to FIG. 3, features from the vehicle control system **30** are shown in more detail. In particular, FIG. 3 shows a flow chart for determining the tracking and prediction of objects based on real time, online, sensor data. By online, we mean when the AV **10** is active in its field, or environment.

[0045] The sensors **12**, including the camera **12_2** and the radar **12_4**, may also include a laser **12_5** and an imaging radar **12_6**.

[0046] The camera **12_2** images may be used to generate a junction state **50** in combination with a shared map **52**. A junction may be defined as an area whose access state changes dynamically. By access state we mean whether the AV **10** is able to access to area being controlled. Examples of junctions include traffic lights, pelican crossings, zebra crossings, give-way markings, etc.

[0047] The camera **12_2**, laser **12_5**, and imaging radar **12_6** data, may be used to generate a semantic vision **54**. The term semantic used herein may generally refer to something in the form of a class or an embedding. Semantic vision may include embeddings of objects viewed in the visual, camera image, modality.

[0048] The laser **12_5** may be used to generate a semantic laser **56**. This may mean an embedding associated with laser points in the laser domain.

[0049] The laser may also be used to generate a model-free laser **58**. The term model-free may generally be used herein to mean raw data points from a particular modality. The model-free laser may be raw laser points.

[0050] The imaging radar **12_6** may be used to generate a semantic radar. The semantic radar **60** may be embeddings defining features captured by the imaging radar **12_6**. As used herein, the term semantic may mean a first representation processed into a second representation bearing semantic information.

[0051] The radar **12_5** may be used to generate a model-free radar **62**. The model-free radar may include raw radar points.

[0052] The junction state **50**, semantic vision **54**, semantic laser **56**, model-free laser **58**, semantic radar **60**, and model-free radar **62**, may be used to generate the tracking and prediction module **38**.

[0053] FIG. 4 shows a similar diagram to FIG. 3, except the inputs are simulated inputs. In this way, the tracking and prediction module **38** uses data from a simulated camera **12_2'**, a simulated laser **12_5'**, a simulated imaging radar **12_6'**, and a simulated radar **12_4'**.

[0054] With reference to FIG. 5, a flow chart is shown to explain how to train an encoder to generate a scenario encoding.

[0055] The tracking and prediction module **38** has determined positions of static objects and positions and trajectories of dynamic objects, which are input together with object and freespace data (derived from model-free data) into a scenario encoder **70**. The term freespace is used to mean a travel envelope which avoids infractions. The travel envelope may be in space time. Infractions represent events that either reduce the safety of operation or reduce the perceived quality of operation. Infractions may include collisions and harsh braking, for example.

[0056] The scenario encoder **70** may receive a further input from a field bug library **72**. The field bug library **72** is a database which includes recorded responses of the AV **10** to infractions online. The scenario encoder **70** may be an autoencoder. The autoencoder may be a variational autoencoder (VAE). A multivariable distribution may be extracted at a bottleneck of the VAE and used as a scenario encoding **74**.

[0057] The scenario encoding **74** may be stored in a scenario library **76**.

[0058] A scenario comparator **78** may also be provided. The scenario comparator **78** may be configured to compare the scenario encoding **74** to the other known scenarios included in the scenario library **76**. The scenario encoder **70** may be trained in a supervised manner to generate scenario encodings **74** which may match known scenarios of the same scene, for example.

[0059] FIG. 6 shows a flow chart explaining how additional scenarios are encoded online using the scenario encoder **70** trained by the method outlined with reference to FIG. 5.

[0060] Online, the dynamic objects, static objects (or static map), and freespace observations, are combined into a bird's eye view **80**. The bird's eye view **80** may be a representation of a scene from a reference point. The reference point may be from above. Additionally, the bird's eye view may be constructed entirely from spatio-temporal sensor readings from the AV itself, e.g. single "frame" LIDAR/RADAR scans or multiple scans integrated over time.

[0061] The bird's eye view **80** may be used as an input to the learned scenario encoder **70**, which generates a scenario encoding **74**. The scenario encoding may be compared with known scenarios from the scenario library **76** using the scenario comparator **78**.

[0062] With reference to FIG. 7, the scenario comparator **78** is a matcher. The matcher may include a plurality of matching means. For example, a dynamic time warping (DTW) matcher **82** may be used to compare points from each compared encoding in time.

[0063] And output from the DTW matcher **82** may be input to a sub-component matcher **84** which distributes the paired points from the DTW matcher **82** to the plurality of matching means. The plurality of matching means each computes a distance between respective points in the pairs.

[0064] The matching means may comprises a Euclidian distance matcher **86**, a learned distance matcher **88** and a custom matcher **90**. The Euclidian distance matcher **86** calculates a Euclidian distance between points of compared encodings. The learned distance matcher **88** may be a neural network trained using contrastive learning to compute a distance between points of compared encodings. The custom matcher may calculate the distance by comparing a hybrid weighted similarity between different heuristics.

[0065] The output of the respective matching means is a match score **92A**, **92B**, **92C**. The match score may be a numerical value. A respective score weight **94A**, **94B**, **94C**, may be applied to each match score. The term "apply" may be used to mean multiply. A summation block **96** may add the weighted match scores to generate a final score **98**. The score weights may be different. The score weights may be predetermined. In other embodiments, the score weights may be learned. For example, the score weights may be adjusted using an optimisation algorithm on training samples having a desired final score for a particular encoding pair.

[0066] With reference to FIG. 8, there is provided a flow chart outlining a computer-implemented method of determining a manoeuvring constraint of an autonomous vehicle during a known scenario according to one or more embodiments.

[0067] At step **100**, the method comprises retrieving a descriptor of the known scenario from the scenario library **76** stored in a database, the descriptor being a multivariate distribution from a bottleneck of a variational autoencoder as described above. Also at step **100**, the descriptor is decoded by a decoder. The decoder may be from the variational autoencoder used to generate the known scenario.

[0068] At step **102**, the method comprises running the known scenario of the AV as a simulation in a simulator (SIM). The SIM is run iteratively by applying different dynamic parameters to the AV. The dynamic parameters may be selected from a list including velocity, acceleration, deceleration, and jerk.

[0069] A first set of dynamic parameters may be applied to the AV in the known scenario. For example, the AV may be travelling at 20 miles per hour (mph) with zero acceleration. At step **104**, safe operation of the AV is checked. Safe operation may be understood as meaning that there are no infractions. An infraction may be an event which adversely affects the user's perception of safe driving. Infractions may include collisions with objections within a scenario. Infractions may also include near misses, e.g. where the AV comes within a predetermined distance of another object in the scenario. These examples are non-exhaustive and are included for illustrative purposes only.

[0070] If there are infractions, at step **106**, the method comprises checking if the dynamic parameters are at their limits. In other words, there is a check to determine if modulation of the dynamic parameters has been exhausted. For example, if the current speed is 20 mph, it is possible to decrement the speed. The decremented (or increments) may be at a predefined interval. The

decrements may be at an interval of 5 mph.

[0071] At step **108**, the method comprises modulating the dynamic parameters. In the above example, the new iteration of speed may be 15 mph.

[0072] At step **104**, safe operation of the AV is again checked at the new speed iteration, e.g. 15 mph.

[0073] The method continues by iteratively running the simulation by incrementing and/or decrementing the dynamic parameters; and checking for infractions at each increment and/or decrement. The method continues incrementing and/or decrementing the dynamic parameters until modulation of the dynamic parameters has been exhausted. For example, the dynamic parameter speed may be exhausted when the speed of the AV is 5 mph.

[0074] When step **104** determines that the operation is safe, the maximum dynamic parameter of the AV below which no infractions occur, and above which an infraction occurs, during the SIM, is identified at step **110**.

[0075] At step **112**, the maximum dynamic parameter is stored as a manoeuvring constraint of the AV for the known scenario, in a constraint library. A manoeuvring constraint may comprise a travel envelope constructed in space-time (e.g. using a space-time graph), within which no infractions are predicted to occur.

[0076] Otherwise, where the modulation has been exhausted at step **106**, the minimum dynamic parameter may be determined as a trigger for initiating a minimal risk manoeuvre (MRM). A minimal risk manoeuvre may include an emergency brake event or pulling to the side of the road, for example. These MR examples are non-exhaustive and are included here for illustrative purposes only.

[0077] At step **114**, the MRM trigger is identified.

[0078] At step **112**, the MRM is stored in the constraint library.

[0079] With reference to FIG. **9**, there is provided a computer-implemented method of controlling an autonomous vehicle according to one or more embodiments.

[0080] The method comprises, at step **120**, encoding the current scenario using the encoder.

[0081] At step **122**, the encoding is obtained from the encoded scenario.

[0082] At step **124**, the encoding of the current scenario is compared with known scenarios from the scenario library. The comparison may be made using the matcher from FIG. **7**. In this way, determining the distance comprises calculating a distance using a plurality of weighted matchers. The comparisons may be exhaustive in that all known scenarios are compared. In other embodiments, the known scenarios may be clustered in the scenario library using an unsupervised clustering model to group similar scenarios together. In this way, fewer comparisons need to be made.

[0083] As described above, the current scenario and each known scenario may be a descriptor representing a multivariable distribution from a bottleneck of a variational autoencoder.

[0084] At step **126**, the distance between the current scenario and a closest known scenario from the database of known scenarios (scenario library) is determined.

[0085] At step **128**, a safety orchestrator is used to determine a manoeuvre to be carried out by the AV. The safety orchestrator retrieves the manoeuvring constraint associated with the closest known scenario from the constraint library from FIG. **8**.

[0086] The safety orchestrator may decide it is best to proceed to step **130** and control the AV by manoeuvring the autonomous vehicle within a manoeuvring constraint associated with the closest scenario when the distance is below a first threshold. In other words, the manoeuvring constraint of the closest scenario may be used as the manoeuvring constraint for the current scenario. The manoeuvring constraint may comprise a travel envelope constructed in space-time, within which no infractions are predicted to occur.

[0087] The safety orchestrator may decide it is best to control the AV at step **132** by performing a minimal risk manoeuvre when the distance is above a second threshold. The second threshold is

greater than the first threshold. The minimal risk manoeuvre may comprise one or more of an emergency stop and pulling over to a road side.

[0088] The safety orchestrator may decide it is best to interpolate, again at step **130**, the manoeuvring constraint associated with the closest scenario when the distance is between the first threshold and the second threshold and manoeuvre the autonomous vehicle within the interpolated manoeuvring constraint.

[0089] Once the mode of implementation (e.g. harvesting the manoeuvring constraint from a closest scenario, interpolation, or by MRM) has been decided, at step **134**, a planner determines control instructions for the AV's actuators to implement the mode of implementation.

Simultaneously, a watchdog monitors the planner's actions. In certain circumstances, the watchdog may override the planner.

[0090] Whilst the foregoing embodiments have been described to illustrate the subject-matter of the present disclosure, the features of the embodiments are not to be taken as limiting the scope of protection. For the avoidance of doubt, the scope of protection is defined by the following claims.

Claims

1. A computer-implemented method of controlling an autonomous vehicle, the method comprising: receiving current sensor data defining a current scenario; determining a distance between the current scenario and a closest scenario from a database of known scenarios; and controlling the autonomous vehicle by: maneuvering the autonomous vehicle within a maneuvering constraint associated with the closest scenario when the distance is below a first threshold; performing a minimal risk maneuver when the distance is above a second threshold, wherein the second threshold is greater than the first threshold; and interpolating the maneuvering constraint associated with the closest scenario when the distance is between the first threshold and the second threshold, and maneuvering the autonomous vehicle within the interpolated maneuvering constraint.
2. The computer-implemented method of claim 1, wherein the current scenario and each known scenario is a descriptor representing a multivariable distribution from a bottleneck of a variational autoencoder.
3. The computer-implemented method of claim 1, wherein determining the distance comprises calculating a distance using a plurality of weighted matchers.
4. The computer-implemented method of claim 3, wherein the plurality of weighted matchers comprises one or more of: a time warping matcher, a Euclidian distance matcher, a cosine distance matcher, and a learned matcher.
5. The computer-implemented method of claim 1, wherein the maneuvering constraint comprises a travel envelope constructed in space-time, within which no infractions are predicted to occur.
6. The computer-implemented method of claim 1, wherein the minimal risk maneuver comprises one or more of an emergency stop and pulling over to a road side.
7. A computer-implemented method of determining a maneuvering constraint of an autonomous vehicle during a known scenario, the method comprising: running a known scenario of the autonomous vehicle as a simulation in a simulator; iteratively running the simulation by applying different dynamic parameters to the autonomous vehicle; identifying a maximum dynamic parameter from the different dynamic parameters applied to the autonomous vehicle below which no infractions occur during the simulation and above which an infraction occurs during the simulation; and storing, in a constraint database, the maximum dynamic parameter as a maneuvering constraint of the autonomous vehicle for the known scenario.
8. The computer-implemented method of claim 7, further comprising: retrieving a descriptor of the known scenario from a database, the descriptor being a multivariate distribution from a bottleneck of a variational autoencoder; and decoding the descriptor using a decoder from the variational autoencoder to generate the known scenario.

- 9.** The computer-implemented method of claim 7, wherein iteratively running the simulation comprises: incrementing the dynamic parameters; and checking for infractions at each increment.
- 10.** The computer-implemented method of claim 9, further comprising: continuing the incrementing until modulation of the dynamic parameters has been exhausted.
- 11.** The computer-implemented method of claim 9, wherein the incrementing comprises incrementing at a predefined interval.
- 12.** The computer-implemented method of claim 7, wherein the dynamic parameters are selected from a list including velocity, acceleration, deceleration, and jerk.
- 13.** The computer-implemented method of claim 7, wherein the maneuvering constraint comprises a travel envelope constructed in space-time, within which no infractions are predicted to occur.
- 14.** A transitory, or non-transitory, computer-readable medium having instructions stored thereon that, when executed by a processor, cause the processor to perform the computer-implemented method of claim 1.
- 15.** The computer-implemented method of claim 7 wherein iteratively running the simulation comprises: decrementing the dynamic parameters; and checking for infractions at each decrement.
- 16.** The computer-implemented method of claim 15, further comprising: continuing the decrementing until modulation of the dynamic parameters has been exhausted.
- 17.** The computer-implemented method of claim 15, wherein the decrementing comprises decrementing at a predefined interval.
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